

National Neighborhood Data Archive (NaNDA): Arts, Entertainment, and Recreation



Organizations by Census Tract, United States, 2003-2017

openICPSR-115543

nanda_artsentrec_tract_2003-2017_01P.dta

nanda_artsentrec_tract_2003-2017_01P.csv

nanda_artsentre0317T_01P.sas7bdat

Overview and Data Dictionary

Documentation Version: 1.1

Last updated: 12/7/2020

Dataset Overview

Description

This dataset contains measures of the number and per capita density of select types of arts, entertainment, and recreation organizations—such as museums, libraries, spectator sports organizations, amusement parks, fitness centers, bowling alleys, and casinos—per United States census tract from 2003 through 2017.

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Funding Sources

- United States Department of Health and Human Services. Administration for Community Living. National Institute on Disability, Independent Living, and Rehabilitation Research (90RTHF0001)
- United States Department of Health and Human Services. National Institutes of Health. National Institute on Aging (RF1-AG-057540)

Data Sources

The North American Industry Classification System (NAICS) was used to identify arts, entertainment, and recreation organizations. NAICS was developed by the United States Office of Management and Budget and is “the standard used by Federal statistical agencies in classifying business establishments for the purpose of collecting, analyzing, and publishing statistical data related to the U.S. business economy” (United States Census Bureau, 2017).

Establishment data was taken from the National Establishment Time Series (NETS) database, which provides annual records of the US economy for more than 60 million private for-profit and nonprofit establishments and government agencies (Walls & Associates, 2017). This longitudinal data source classifies establishments by NAICS code and provides detailed business microdata and address history. NETS is produced in most years by Walls and Associates using archival establishment data from Dun & Bradstreet. While it currently includes data from 1990 to 2015 and 2017, only data from 2003-2015 and 2017 were used to create this dataset. (NETS data was not produced for 2016.)

Census tracts for establishments were determined using the 2010 TIGER/Line shapefiles produced by the United States Census Bureau (United States Census Bureau, 2010).

Census tract population figures for 2000-2010 were taken from Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010 (Clarke & Melendez, 2019). Population figures for 2011-2017 were obtained from Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017 (Melendez et al., 2020).

Coverage

The dataset contains one observation per census tract per year in the United States, including Puerto Rico and the US Virgin Islands. Other US island territories are excluded.

Methodology

This dataset highlights the concentration of gyms, libraries, bowling alleys, and other arts, entertainment, and recreation organizations per census tract in the United States. These types of organizations—along with other gathering places such as senior centers, churches, coffee shops, and retail stores—function as common “third places” in the United States (Finlay et al.,

2018). Third places are “physical locations outside of the home (first place) or work/school (second place) that facilitate social interaction, community building, and social support” (Finlay et al., 2019). They are social infrastructure vital to civic life (Klinenberg, 2018).

Our research team identified the following NAICS codes of interest:

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Note that NAICS codes are hierarchical, and three-digit codes are inclusive of all six-digit codes that start with the same pattern. For example, if a tract contains two amusement and recreation centers (7139), one gym (71394), and one bowling alley (713950), the total number of organizations is two, not four.

For each NAICS code we identified, we queried the NETS database for all establishments that were open at some time between 2003 and 2015 and in 2017 (NETS does not contain data for 2016), and their location during each year. Note that NETS data were not cleaned or verified during this process, and that NETS data has limitations and inaccuracies as described in Gomez-Lopez et al. (2017) and Finlay et al. (2019).

We determined each establishment’s census tract for each year by mapping its latitude and longitude provided in the NETS database to the 2010 version of the TIGER/Line shapefiles from the US Census Bureau.

Using these locations, we created three variables to represent establishment counts per tract. The first count variable is the total of all establishments with that NAICS code in the tract. The second is the count of just establishments with non-zero sales figures in the current year, which can be used where desired to filter out establishments with very low business volume. The third is the count of establishments employing more than one person, which can be used to filter out very small owner-operated businesses if desired. The most useful variable for a given research question may vary by establishment type, so we have created three count types for all NAICS codes so that users may make their own decision about the version that best meets their needs.

We then calculated population densities (per 1000 people) for each of the three count variables for each NAICS code. Tract populations for 2003-2009 come from the NaNDA dataset [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#). Values for each year are interpolated. For 2010-2012 and 2013-2017, we used the tract population from the NaNDA dataset [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#). Population figures in this dataset were taken from the 2012 and 2017 ACS five-year estimates, respectively, in order to avoid overlapping ACS estimates as advised by the Census Bureau, and for compatibility with other NaNDA datasets. Population figures for 2010-2017 are not interpolated. Consult the codebooks for the NaNDA datasets [Neighborhood Socioeconomic and Demographic Characteristics of Census Tracts, United States, 2000-2010](#) (Clarke & Melendez, 2019) and [Socioeconomic Status and Demographic Characteristics of Census Tracts, United States, 2008-2017](#) (Melendez et al., 2020) for details on why these population figures were selected and how 2003-2009 population was interpolated.

We also created three variables representing the density of establishments per square mile. The area is the land area of the tract from the 2010 version of the TIGER/Line shapefiles. As a result, all area densities are standardized to 2010 census geography and are comparable across years across the entire dataset.

Population density was set to -1 for tracts with zero population in a given year. Population and area densities are missing for tracts in Puerto Rico and the US Virgin Islands, which are not included in the TIGER/Line shapefiles. Population and area densities are also missing in certain years for a few census tracts whose FIPS codes changed at some point between 2003 and 2017.

Finally, we merged counts and densities of all selected NAICS codes to create an aggregate dataset containing tract population, area, and count and density variables for all arts, entertainment, and recreation organizations per tract in each year.

Usage Note

Arts, entertainment, and recreation organizations are one of many common third places in communities. Data users interested in third places and community resources might find further useful data in these additional NaNDA datasets:

- [Eating and drinking places by census tract, United States, 2003-2017](#) (Esposito et al., 2020), which includes coffee shops, fast food restaurants, and bars.
- [Grocery stores by census tract, United States, 2003-2017](#) (Finlay et al., 2020a).
- [Personal care services and laundromats by census tract, United States, 2003-2017](#) (Finlay et al., 2020c), which includes barber shops and hair and nail salons.
- [Post offices and banks by census tract, United States, 2003-2017](#) (Finlay et al., 2020d).
- [Religious, civic, and social organizations by census tract, United States, 2003-2017](#) (Finlay et al., 2020b).
- [Retail establishments by census tract, United States, 2003-2017](#) (Finlay et al., 2020e), which includes stores selling furniture, clothing, sporting goods, home and garden supplies, books, and used merchandise, among other items.
- [Social services by census tract, United States, 2003-2017](#) (Finlay et al., 2020f), which includes senior centers.

Variables

For additional information on the definition of each NAICS code, see the Methodology section of this document and the 2017 NAICS website: <https://www.census.gov/cgi-bin/sssd/naics/naicsrch?chart=2017>.

Variable	Type	Obs	Unique	Mean	Min	Max	Label
tract_fips10	string	1036462	74033	.	.	.	Census tract FIPS code (2010 TIGER/Line shapefiles)
year	int	1036462	14	2009.571	2003	2017	Year
population	float	1020881	250271	4208.894	0	65528	Tract population
aland10	float	1022434	72534	48.07265	0	85425.73	Tract land area, square miles
count_7111	int	1036462	108	0.8826653	0	170	# performing arts orgs
count_sales_7111	int	1036462	98	0.7687527	0	154	# performing arts orgs w/ sales>0
count_emp_7111	int	1036462	45	0.3084532	0	73	# performing arts orgs w/ 2+ employees
popden_7111	float	1015306	156078	1.230072	-1	289507.3	# performing arts orgs per 1000 people
popden_sales_7111	float	1015280	144353	1.150701	-1	289507.3	# performing arts orgs per 1000 people w/ sales>0
popden_emp_7111	float	1015171	89405	0.1954052	-1	17744.44	# performing arts orgs per 1000 people w/ 2+ employees
aden_7111	float	1017996	108686	1.717794	0	1117.016	# performing arts orgs per sq mile
aden_sales_7111	float	1017996	104203	1.496598	0	1071.884	# performing arts orgs per sq mile w/ sales>0
aden_emp_7111	float	1017996	70191	0.6184127	0	471.6009	# performing arts orgs per sq mile w/ 2+ employees
count_7112	int	1036462	29	0.3095164	0	120	# spectator sports orgs
count_sales_7112	int	1036462	27	0.2821155	0	120	# spectator sports orgs w/ sales>0
count_emp_7112	int	1036462	26	0.2488147	0	118	# spectator sports orgs w/ 2+ employees
popden_7112	float	1015273	88536	0.2070368	-1	13234.82	# spectator sports orgs per 1000 people
popden_sales_7112	float	1015262	83684	0.1902338	-1	13234.82	# spectator sports orgs per 1000 people w/ sales>0
popden_emp_7112	float	1015268	75027	0.1897253	-1	13234.82	# spectator sports orgs per 1000 people w/ 2+ employees
aden_7112	float	1017996	40450	0.2678594	0	312.0844	# spectator sports orgs per sq mile
aden_sales_7112	float	1017996	39875	0.2424507	0	312.0844	# spectator sports orgs per sq mile w/ sales>0
aden_emp_7112	float	1017996	36854	0.216394	0	312.0844	# spectator sports orgs per sq mile w/ 2+ employees
count_712	int	1036462	74	0.4535584	0	161	# museums/historical sites

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
count_sales_712	int	1036462	71	0.4166733	0	151	# museums/historical sites w/ sales>0
count_emp_712	int	1036462	67	0.3757697	0	150	# museums/historical sites w/ 2+ employees
popden_712	float	1015388	104714	0.5903121	-1	98649.76	# museums/historical sites per 1000 people
popden_sales_712	float	1015370	99021	0.5740767	-1	98649.76	# museums/historical sites per 1000 people w/ sales>0
popden_emp_712	float	1015361	90772	0.3087538	-1	19384.52	# museums/historical sites per 1000 people w/ 2+ employees
aden_712	float	1017996	48617	0.69355	0	597.6282	# museums/historical sites per sq mile
aden_sales_712	float	1017996	47661	0.6299637	0	564.4267	# museums/historical sites per sq mile w/ sales>0
aden_emp_712	float	1017996	45216	0.5865673	0	597.6282	# museums/historical sites per sq mile w/ 2+ employees
count_51912	int	1036462	24	0.39995	0	23	# libraries/archives
count_sales_51912	int	1036462	22	0.3837275	0	21	# libraries/archives w/ sales>0
count_emp_51912	int	1036462	23	0.3623654	0	22	# libraries/archives w/ 2+ employees
popden_51912	float	1015202	107172	0.3165315	-1	29076.78	# libraries/archives per 1000 people
popden_sales_51912	float	1015198	104852	0.3107341	-1	29076.78	# libraries/archives per 1000 people w/ sales>0
popden_emp_51912	float	1015195	99971	0.3044587	-1	29076.78	# libraries/archives per 1000 people w/ 2+ employees
aden_51912	float	1017996	35385	0.4159464	0	187.9288	# libraries/archives per sq mile
aden_sales_51912	float	1017996	35101	0.3971806	0	187.9288	# libraries/archives per sq mile w/ sales>0
aden_emp_51912	float	1017996	34230	0.3886159	0	165.2535	# libraries/archives per sq mile w/ 2+ employees
count_7131	int	1036462	15	0.0678462	0	14	# amusement parks/arcades
count_sales_7131	int	1036462	15	0.0618383	0	14	# amusement parks/arcades w/ sales>0
count_emp_7131	int	1036462	14	0.0595565	0	13	# amusement parks/arcades w/ 2+ employees
popden_7131	float	1015157	29457	0.1190794	-1	22363.54	# amusement parks/arcades per 1000 people
popden_sales_7131	float	1015151	27402	0.1052646	-1	13234.82	# amusement parks/arcades per 1000 people w/ sales>0
popden_emp_7131	float	1015147	26246	0.0883776	-1	13234.82	# amusement parks/arcades per 1000 people w/ 2+ employees
aden_7131	float	1017996	9653	0.0605362	0	73.72456	# amusement parks/arcades per sq mile
aden_sales_7131	float	1017996	9482	0.0547324	0	73.72456	# amusement parks/arcades per sq mile w/ sales>0
aden_emp_7131	float	1017996	9084	0.0534523	0	73.72456	# amusement parks/arcades per sq mile w/ 2+ employees
count_7132	int	1036462	20	0.0450185	0	21	# casinos
count_sales_7132	int	1036462	20	0.041853	0	21	# casinos w/ sales>0
count_emp_7132	int	1036462	19	0.0351282	0	21	# casinos w/ 2+ employees
popden_7132	float	1015067	22288	0.0172158	-1	666.6667	# casinos per 1000 people

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Variable	Type	Obs	Unique	Mean	Min	Max	Label
popden_sales_7132	float	1015065	20918	0.0163055	-1	666.6667	# casinos per 1000 people w/ sales>0
popden_emp_7132	float	1015067	18215	0.0143702	-1	666.6667	# casinos per 1000 people w/ 2+ employees
aden_7132	float	1017996	5932	0.0428801	0	50.99405	# casinos per sq mile
aden_sales_7132	float	1017996	5886	0.0398378	0	50.99405	# casinos per sq mile w/ sales>0
aden_emp_7132	float	1017996	5446	0.0326769	0	50.99405	# casinos per sq mile w/ 2+ employees
count_721120	int	1036462	44	0.0211112	0	88	# casino hotels
count_sales_721120	int	1036462	41	0.0195926	0	85	# casino hotels w/ sales>0
count_emp_721120	int	1036462	44	0.0200249	0	82	# casino hotels w/ 2+ employees
popden_721120	float	1015050	7991	0.0104421	-1	666.6667	# casino hotels per 1000 people
popden_sales_721120	float	1015050	7488	0.0095265	-1	666.6667	# casino hotels per 1000 people w/ sales>0
popden_emp_721120	float	1015050	7647	0.0100721	-1	666.6667	# casino hotels per 1000 people w/ 2+ employees
aden_721120	float	1017996	2183	0.0167857	0	66.40314	# casino hotels per sq mile
aden_sales_721120	float	1017996	2151	0.0155369	0	66.40314	# casino hotels per sq mile w/ sales>0
aden_emp_721120	float	1017996	2140	0.0157555	0	66.40314	# casino hotels per sq mile w/ 2+ employees
count_7139	int	1036462	75	3.027258	0	104	# all golf, skiing, boating, fitness, bowling, other
count_sales_7139	int	1036462	74	2.748857	0	91	# all golf, skiing, boating, fitness, bowling, other w/ sales>0
count_emp_7139	int	1036462	73	2.518073	0	88	# all golf, skiing, boating, fitness, bowling, other w/ 2+ employees
popden_7139	float	1015953	307192	2.543786	-1	507204.4	# all golf, skiing, boating, fitness, bowling, other per 1000 people
popden_sales_7139	float	1015921	296532	1.582651	-1	55908.86	# all golf, skiing, boating, fitness, bowling, other per 1000 people w/ sales>0
popden_emp_7139	float	1015879	284155	2.26767	-1	507204.4	# all golf, skiing, boating, fitness, bowling, other per 1000 people w/ 2+ employees
aden_7139	float	1017996	235870	3.019359	0	601.394	# all golf, skiing, boating, fitness, bowling, other per sq mile
aden_sales_7139	float	1017996	230303	2.696968	0	555.1329	# all golf, skiing, boating, fitness, bowling, other per sq mile w/ sales>0
aden_emp_7139	float	1017996	222320	2.528954	0	555.1329	# all golf, skiing, boating, fitness, bowling, other per sq mile w/ 2+ employees
count_71394	int	1036462	60	1.163099	0	83	# gyms/fitness centers
count_sales_71394	int	1036462	53	1.020672	0	81	# gyms/fitness centers w/ sales>0
count_emp_71394	int	1036462	59	1.057616	0	83	# gyms/fitness centers w/ 2+ employees
popden_71394	float	1015353	184143	1.289392	-1	507204.4	# gyms/fitness centers per 1000 people

National Neighborhood Data Archive (NaNDA)
Arts, Entertainment, and Recreation Organizations by Census Tract, United States, 2003-2017

Variable	Type	Obs	Unique	Mean	Min	Max	Label
popden_sales_71394	float	1015327	170561	0.4460681	-1	39704.46	# gyms/fitness centers per 1000 people w/ sales>0
popden_emp_71394	float	1015338	172567	1.256578	-1	507204.4	# gyms/fitness centers per 1000 people w/ 2+ employees
aden_71394	float	1017996	132324	1.390792	0	478.0311	# gyms/fitness centers per sq mile
aden_sales_71394	float	1017996	127534	1.215148	0	447.1904	# gyms/fitness centers per sq mile w/ sales>0
aden_emp_71394	float	1017996	127676	1.277995	0	462.6108	# gyms/fitness centers per sq mile w/ 2+ employees
count_713910	int	1036462	14	0.2333863	0	13	# golf courses
count_sales_713910	int	1036462	13	0.2280662	0	12	# golf courses w/ sales>0
count_emp_713910	int	1036462	14	0.21968	0	13	# golf courses w/ 2+ employees
popden_713910	float	1015254	81249	0.1607129	-1	16628.21	# golf courses per 1000 people
popden_sales_713910	float	1015250	80048	0.1551828	-1	16628.21	# golf courses per 1000 people w/ sales>0
popden_emp_713910	float	1015245	77834	0.1569968	-1	16628.21	# golf courses per 1000 people w/ 2+ employees
aden_713910	float	1017996	19980	0.0761456	0	55.91029	# golf courses per sq mile
aden_sales_713910	float	1017996	19857	0.0734003	0	55.91029	# golf courses per sq mile w/ sales>0
aden_emp_713910	float	1017996	19392	0.0709403	0	55.91029	# golf courses per sq mile w/ 2+ employees
count_713950	int	1036462	17	0.0995164	0	17	# bowling alleys
count_sales_713950	int	1036462	16	0.0957363	0	15	# bowling alleys w/ sales>0
count_emp_713950	int	1036462	14	0.0876829	0	16	# bowling alleys w/ 2+ employees
popden_713950	float	1015065	43244	0.0331634	-1	500	# bowling alleys per 1000 people
popden_sales_713950	float	1015064	42110	0.0319139	-1	500	# bowling alleys per 1000 people w/ sales>0
popden_emp_713950	float	1015065	39186	0.0298174	-1	500	# bowling alleys per 1000 people w/ 2+ employees
aden_713950	float	1017996	10942	0.0672189	0	58.33602	# bowling alleys per sq mile
aden_sales_713950	float	1017996	10857	0.0640575	0	58.33602	# bowling alleys per sq mile w/ sales>0
aden_emp_713950	float	1017996	10075	0.0606117	0	58.33602	# bowling alleys per sq mile w/ 2+ employees

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